Elevator Trim Tab System – General Data

1. General

An elevator trim tab, attached to the left elevator trailing edge, is operated by a handwheel which is part of the trim control assembly mounted in the trim console to the right of the pilot's seat.

2. Description

The elevator trim tab system consists of a trim tab, trim handwheel, trim screw jack and trim cables. The cables, connected to a drum attached to the trim handwheel, are routed through pulleys and fairleads up the forward face of the sloping bulkhead, along the cabin roof to the rear fuselage, to connect to a cable drum attached to the screw jack drive shaft.

A. Trim Tab

The elevator trim tab is hinged to the trailing edge of the left elevator. A bracket on the lower surface of the tab provides for the attachment of an adjustable connecting rod which connects to the trim screw jack. Drain holes are provided in the tab bottom skin.

B. Trim Tab Handwheel

The elevator trim tab handwheel is mounted in the trim console, a cable drum to which the trim cables connect, is attached to the handwheel. Mounted on the inboard end of the cable drum is a spiral grooved plate, which through a pin and lever arrangement, drives an indicator pointer.

C. Trim Tab Screw Jack

The elevator trim tab screw jack, mounted on the front spar of the left elevator, converts rotary motion of the cable drum into linear movement of the tab connecting rod attachment. The cable drum is rotated by the trim cables secured to the drum.

3. Operation

Rotary movement of the elevator trim handwheel turns the cable drum, which through the cables turns the screw jack cable drum to extend or retract the screw jack. The screw jack operates the connecting rod, to either raise or lower the trim tab. The trim tab moves up for a nose down selection, or down for a nose up selection of the trim tab handwheel.

4. Removal/Installation

A. Remove Elevator Trim Tab System Rear Pulley Assembly

1) Release elevator trim tab system cables.

2) Remove rear pulley bracket from elevator torque tube.

B. Install Elevator Trim Tab System Rear Pulley Assembly

1) Install rear pulley bracket assembly, referring to Figure 1 for assembly details. Lubricate pulley bush (refer to 12-20-10, Lubrication – General Data). Ensure that after tightening the pivot bolt there is free movement of bracket on bushing.

2) Connect and tension elevator trim tab cables. Rig elevator trim tab system, refer to Para 5.A.

3) Do a function test of the elevator trim tab indicator potentiometer as per 27-33-21, Elevator Trim Position Transmitter, Para 3.B.

4) If adjustments are required to the elevator trim tab indicator potentiometer adjust as per 27-33-21, Elevator Trim Position Transmitter, Para 3.A.

5. Adjustment/Test

A. Rig Elevator Trim Tab System (Mod 6/1219)

Refer to Figure 1 for details.

1) Rotate elevator trim tab handwheel two turns (approximately) from full nose down position until cable terminals are located as shown in Figure 1, “Detail C”, and cables are arranged as in “Detail A”.

2) Maintain handwheel as in Para 5.A.1) and set trim screw jack in left elevator to 0.690 to 0.720 inches extended, and arrange cables as shown in Figure 1, “Detail D”.

3) Maintaining handwheel and screw jack cable drum settings as in Para 5.A.1) and Para 5.A.2), adjust and tension cables in rear fuselage. For cable tension values refer to 27-00-00, Flight Controls – General Data. Lock turnbuckles using locking clips.

4) Disconnect connecting rod between screw jack and tab attachment bracket at tab attachment.

5) Set trim tab in neutral position by aligning trim tab and elevator trailing edges.

6) Maintaining position of screw jack as in Para 5.A.2), adjust and connect trim tab connecting rod.

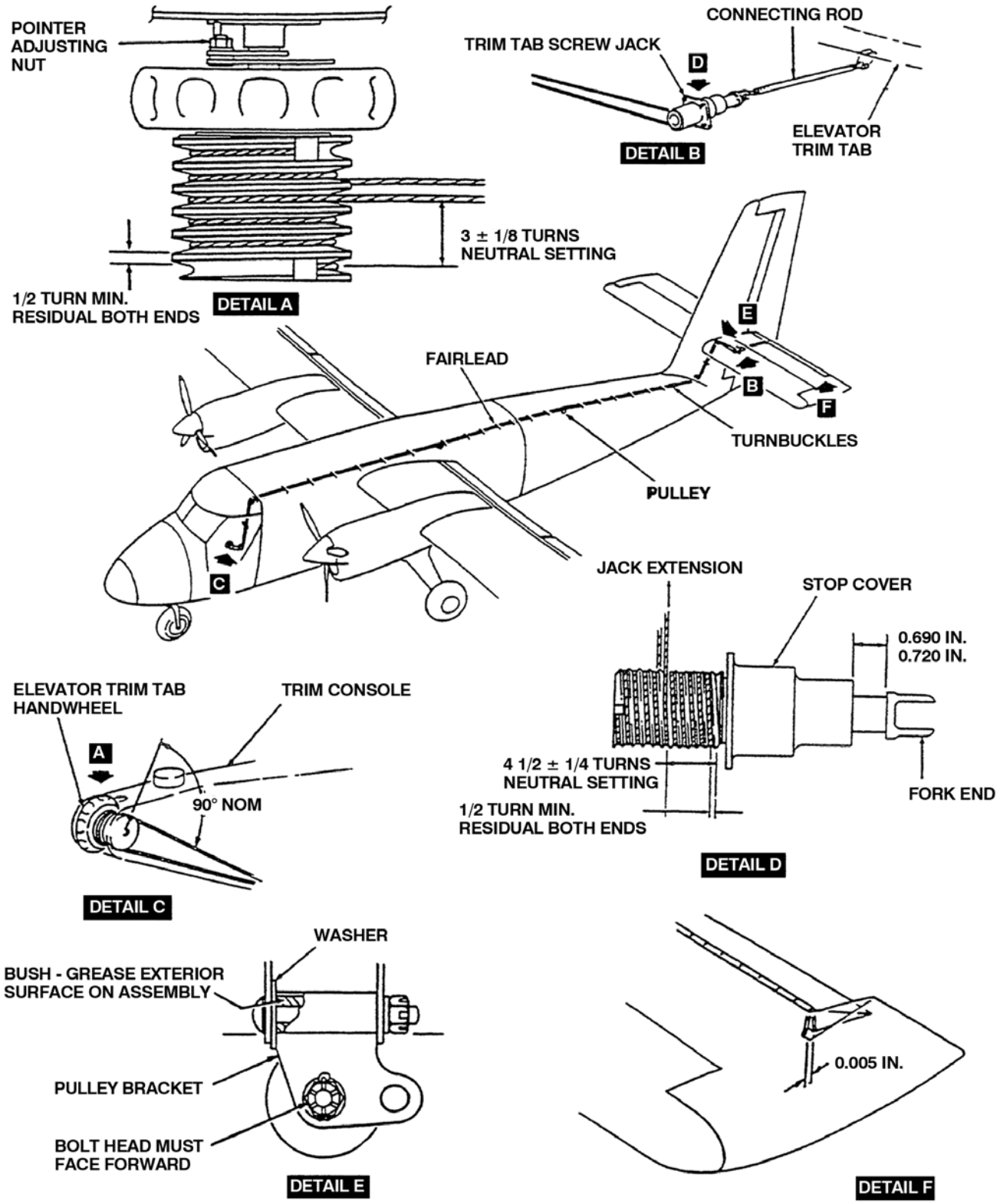
7) Rotate elevator trim handwheel to nose full up position; ensure that screw jack is fully retracted. Check that 0.005 inch exists between trim tab hinge lugs (refer to Figure 1, “Detail F”), if not, lengthen connecting rod to obtain clearance.

8) Tighten and lock connecting rod adjustable end.

9) With elevator trim handwheel still in nose full up position, slacken pointer nut and position pointer to coincide with nose up position. Tighten pointer nut.

10) Check elevator trim tab control surface clearance (refer to 27-00-00, Flight Controls – General Data).

11) Operate trim tab and check for full and free range of movement. Check at extremities of travel that handwheel and screw jack drums have a minimum of half a turn of cable on each drum.



Elevator Trim Tab System – Rigging
Figure 1

6. Inspection/Check

A. Free Play Check – Elevator Trim Tab

Refer to Figure 1 for details.

1) Operate elevators to neutral position.

2) Operate trim tab to neutral position.

3) Move trim tab manually and check that free play (backlash), measured at trailing edge does not exceed 0.10–inch.

Note Measurement to be taken adjacent to control rod attachment point.